

Head and Spine Anatomy and Biomechanics

Colin B. Harris, Rachid Assina, Brandee Gentile, Michael J. Vives

INTRODUCTION

The complexity of the central nervous system (CNS) as well as the bones and soft tissues that provide its protection present special challenges when treating the injured athlete. A thorough understanding of the normal anatomy and biomechanics of the brain and spine are necessary in order to identify pathologic states or unstable injuries and treat them appropriately. The following chapter presents a review of the anatomy of the head and spine, as well as the biomechanics of both normal and injured states. Despite the complexity of this topic, it is possible to identify common patterns that allow an improved understanding of head and spine anatomy that serves as a solid foundation for diagnosing and treating these injuries in the athlete.

ANATOMY

The relationships between the rigid bony structures that surround and protect the brain (skull), the spinal cord, and the nerve roots (cervical, thoracic, lumbar, and sacral-coccygeal vertebrae) as well as the intervening soft tissues that provide dynamic stability are outlined below. Athletic injuries can damage these structures by either sudden, high-energy trauma, such as a direct blow or whiplash type injury, to repeated loading within a more physiologic range that nevertheless can cause pain or disability. Spine and brain injuries are greatly feared by athletes in a variety of sports due to their potential catastrophic nature, such as traumatic brain injury (TBI) and spinal cord injury (SCI), which can have life-altering and often irreversible consequences. The following overview of the relevant anatomy is meant to provide a background for understanding the biomechanics of the head and spine as well as the pathologic states that can lead to these potentially devastating injuries.

SKULL (CRANIUM)

To best understand how concussions and other brain injuries occur, it is important to fully understand and appreciate the anatomy involved. The anatomy covered within this chapter is a brief overview, which begins with the most external parts and then works through internally. The cranium is the skeleton of the head, a union of functional components combined to form a single skeletal formation. The basic functional component is to house and protect the brain. The skull consists of 22 bones;

8 of which protect the brain—4 singular bones centered along the midline (frontal, ethmoidal, sphenoidal, and occipital) and 2 sets of bones occurring as bilateral pairs (temporal and parietal), with the remaining 14 bones comprising an individual's facial structure¹ (Fig. 122.1A). They are essentially fixed, immovable joints, which are connected along multiple suture lines. The only independent movement of the skull occurs at the temporomandibular joint.

BRAIN

The cranial meninges are the covering of the brain that lies immediately within the skull (see Fig. 122.1B). Their function is to protect the brain, provide the blood supply, and provide a space for the flow of cerebral spinal fluid¹. The meninges are composed of three distinct layers: the dura mater, the arachnoid mater, and the pia mater². Each layer has a specific role in the protection of the CNS. The dura mater is the outer-most layer of the meninges and is a tough, thick, bilaminar membrane that adheres to the internal surface of the skull. It is additionally comprised of three layers, which include the periosteal, meningeal, and dural sinus. The periosteal layer lines the skull and the meningeal layer lines the periosteal layer, except when separated by the dural sinus. The dural sinus forms the major venous drainage pathways from the brain, predominantly to the internal jugular veins. The dura mater has an extensive framework of blood supply to itself and provides significant blood supply to the skull through multiple branches of different arteries. The dura mater is also pain sensitive due to being innervated by branches of the trigeminal nerve.^{1,3}

The arachnoid mater is avascular and not attached to the dura mater, but is held in place by the cerebrospinal fluid (CSF). The arachnoid mater is directly above the subarachnoid space, which is a cobweb-like structure housing CSF. The pia mater is the inner most layer of the meninges. It is a highly vascularized structure directly under the subarachnoid space and adheres to the brain by following its contours.

The brain is composed of the cerebrum, cerebellum, and brainstem (see Fig. 122.1D). When the skull and the dura are removed, the gyri, sulci, and fissures are made visible through the arachnoid and pia layers. The cerebrum is what makes up the major portion of the brain and is divided into two major parts: the right and left cerebral hemispheres, which are separated by the great longitudinal fissure. The two hemispheres can then

Abstract

The brain and spinal cord are complex structures that have a limited ability to withstand the external forces athletes may encounter in the course of competition. Head and spine injuries may be catastrophic, leading to permanent cognitive and motor disabilities if not addressed quickly and appropriately. The brain is protected by the meninges and the skull. The surrounding structures that protect the spinal cord from injury include the disc, facet joints, and posterior ligamentous complex. A thorough understanding of the anatomy of the central nervous system and the biomechanics of the spinal motion segment, as well as its response to external loads, are helpful in treating the injured athlete and lead to a greater understanding of both physiologic and pathologic states. Regions of the spine most vulnerable to injury include the transition zones between the cervical and thoracic spine, as well as between the thoracic and lumbar spine, and repetitive micro-trauma can lead to failure even in the absence of a high energy or sudden injury. This chapter provides an overview of the anatomy and biomechanics of the brain and spine with special emphasis on response to external loads and the biomechanics of important stabilizing structures.

Keywords

brain
spine
skull
meninges
disc
ligament
instability

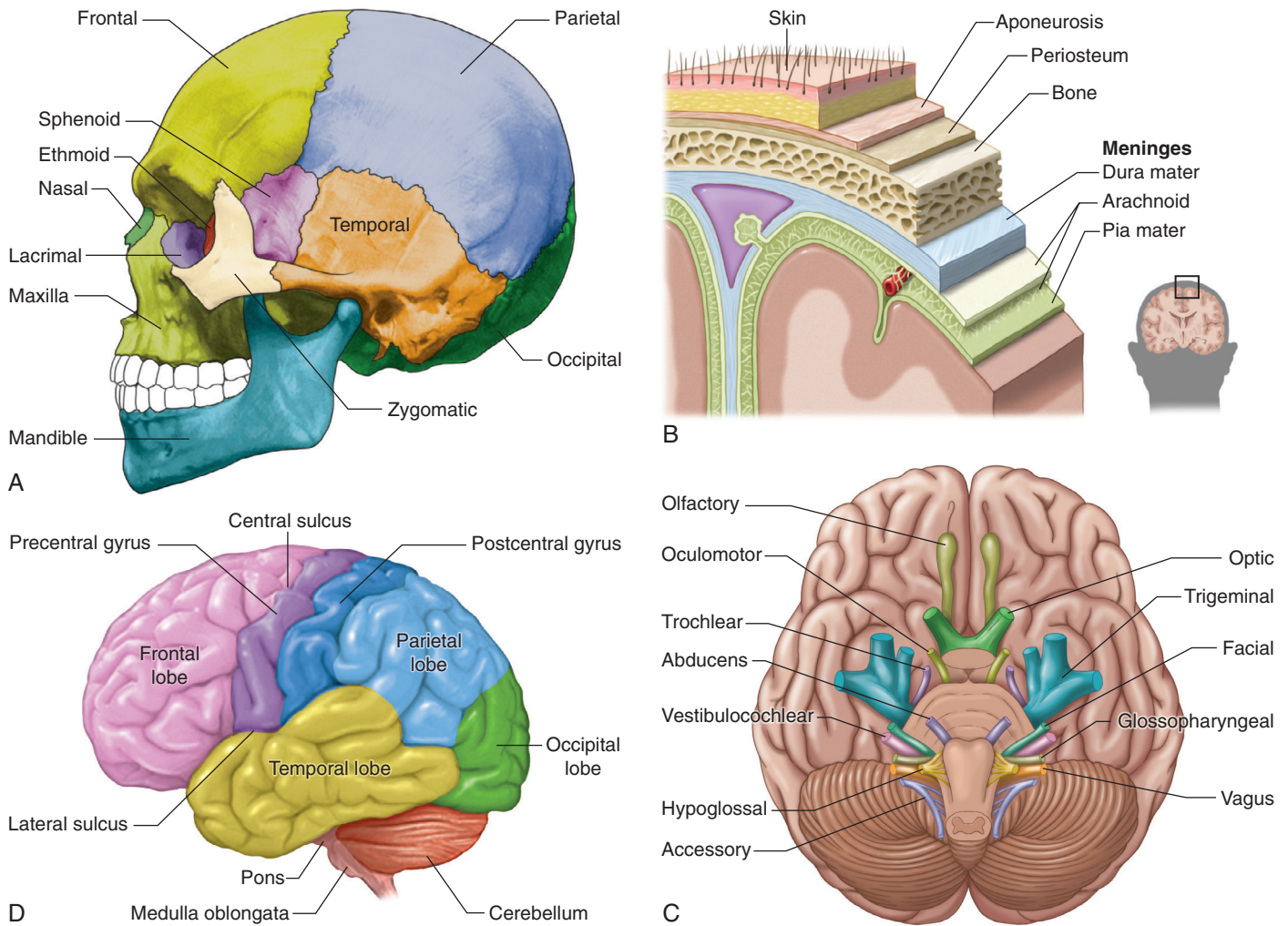


Fig. 122.1 (A) Sagittal view demonstrating the bones of the skull (<https://en.wikipedia.org/wiki/Meninges#/media/File:Meninges-en.svg>). (B) Coronal cross-section demonstrating the layers of the skull and meninges ([https://en.wikipedia.org/wiki/File:Human_skull_side_simplified_\(bones\).svg](https://en.wikipedia.org/wiki/File:Human_skull_side_simplified_(bones).svg)). (C) The brain and brainstem (https://en.wikipedia.org/wiki/Lobes_of_the_brain#/media/File:Blausen_0101_Brain_LateralView.png). (D) View of the inferior surface of the brain demonstrating the cranial nerves (https://en.wikipedia.org/wiki/Cranial_nerves#/media/File:Brain_human_normal_inferior_view_with_labels_en-2.svg).

be subdivided into the frontal, parietal, temporal, and occipital lobes. Each lobe of the brain roughly correlates with the same location as the skull bone by the same name. The two hemispheres of the brain are conjoined by the corpus callosum, which allows the two hemispheres to continually deliver messages to the other. The cerebellum lies posterior to the pons and medulla oblongata. The cerebellum receives afferent information concerning voluntary movement from the cerebral cortex and from the muscles, tendons, and joints.¹ The brainstem consists of the medulla oblongata, pons, and midbrain. This area is crucial for cranial nerve (CN) function, excluding those associated with olfaction and vision, as well as sympathetic and parasympathetic autonomic functions. The brainstem also serves as the location for efferent and afferent pathways to cross between the cerebrum and cerebellum.

The CN are motor or sensory fibers that innervate muscles or glands. An injury can occur to the CN through injury to the skull, excessive movement of the brain, or tumors. There are 12

CN pairs that originate from the brain and penetrate through the dura. They are responsible for a variety of functions and are vulnerable to injury; a detailed review is out of the scope of this chapter (see Fig. 122.1C).

HEAD INJURIES

Head injuries are any injury to the scalp, the skull, or the brain and can range from minor bruising to severe TBI. Head injuries can be classified as closed, in which case the skull remains intact, or open where there is a breakdown of the scalp and/or fracture of the skull. Common head injuries include scalp wounds, skull fractures, concussions, contusions, and hemorrhages.

The skull is composed of outer and inner tables with an intermediate vascular diploic space. Skull fractures occur as a direct head trauma and may involve the cranial vault or the base of the skull. Cranial vault fractures can be open or closed, and can be linear (Fig. 122.2A) or depressed (see Fig. 122.2B). Skull

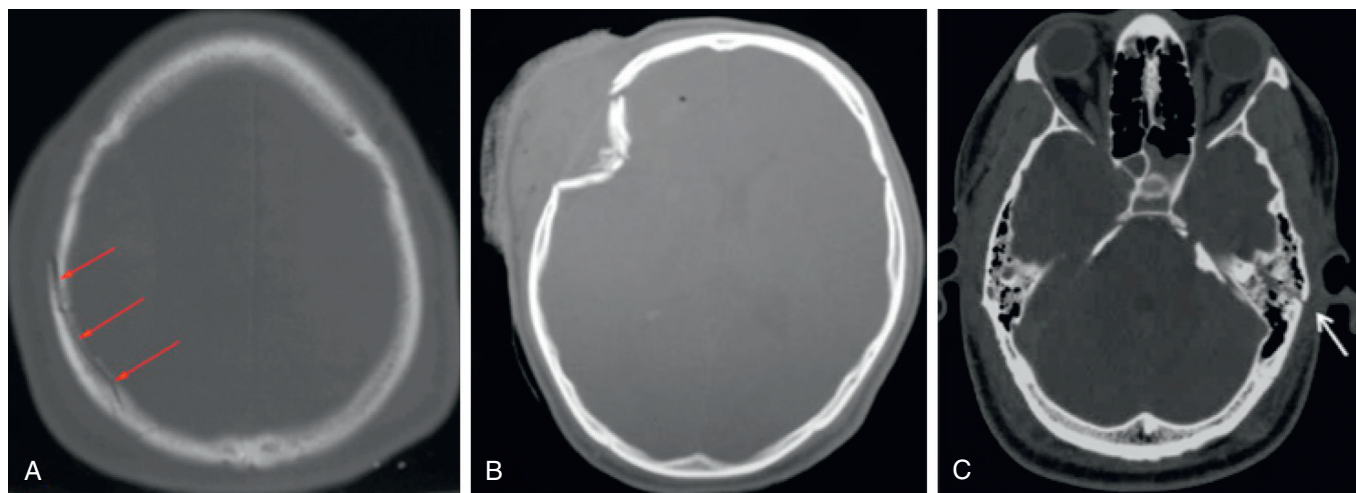


Fig. 122.2 Axial computed tomography images demonstrating a linear cranial vault fracture at the parietal bone as indicated by arrows (A), depressed cranial vault fracture at the frontal bone (B), and linear skull base fracture through the petrous portion of the temporal bone (arrow pointing to left temporal bone illustrates fracture line) (C).

base fractures are typically linear and may be associated with CN damage, dural laceration, and possible CSF leak from the nose (rhinorrhea), or the ear (otorrhea) (see Fig. 122.2C).

Traumatic intracranial hemorrhages occur when a blood vessel within the skull is damaged. Hemorrhage can occur at the epidural, subdural, arachnoid spaces or within the brain matter. A subdural hematoma is the most common hemorrhage after traumatic head injury. It is a brain bleed that has created its own space between the dura mater and the arachnoid mater. It is typically the result of a blow to the head that causes the brain to thrust back and forth within the skull; most often tearing a superior cerebral vein as it enters the superior sagittal sinus. It also can be caused by the rupture of a subdural artery. This type of hemorrhage follows the contour of the brain in a concave shape (Fig. 122.3A). An epidural hematoma is also typically the result of a blow to the head. However, this type of bleed is different from a subdural hematoma in that it is most commonly arterial in origin and occurs when blood from a torn branch of the middle meningeal artery collects between the skull and the dura mater. This can cause compression of brain tissue, requiring surgical evacuation. Epidural hematomas are usually associated with a lucid interval; which is a temporary improvement of the patient's condition after a TBI. The patient lapses into unconsciousness after recovery from the initial concussive force when the continued bleeding causes expansion of the hematoma past the brain's ability to compensate. The hemorrhage expands along the epidural space, but it does not cross the suture lines, forming a concave shape (see Fig. 122.3B). Cerebral contusions are another type of brain bleed that can occur after head trauma, and is a bruise of the brain tissue. It is associated with multiple diffuse or localized microhemorrhages from ruptured veins or arteries. It commonly occurs in coup injuries where the brain is bruised directly under the area of impact, and/or contrecoup injuries in which the brain is injured on the side opposite to the impact. The brain is easily bruised near sharp ridges of the inside of the skull under the frontal and temporal lobes, making the frontal and temporal lobes the most common contused areas

of the brain (see Fig. 122.3D). A subarachnoid hemorrhage is an arterial bleed into the subarachnoid space and may occur as a result of a head injury, the most common, or spontaneously, from a ruptured cerebral aneurysm or an arteriovenous malformation (see Fig. 122.3C).

CONCUSSION

A concussion is a temporary loss of brain function after a head trauma. It is also known as minor head trauma or mild traumatic brain injury. It is the most common type of TBI. The patient may or may not experience a temporary loss of consciousness, and intracranial hemorrhage or skull fracture may or may not be present. Therefore concussion is usually a diagnosis of exclusion and is considered a complex neurobehavioral syndrome with a multi-faceted clinical dilemma. A concussion is caused by a rapid acceleration or deceleration of an individual's brain within their skull, caused by either a direct or indirect blow to their head. This can occur in either a linear or rotational fashion; which then causes intracranial imbalances and metabolic changes to occur.⁴ According to the Centers for Disease Control and Prevention, the number of emergency department, sports recreation-related traumatic brain injury (SRR-TBI) treatments increased between 2001 and 2012. During that time, there were approximately 3.42 million emergency room visits for children between the ages of 0 and 19 for the treatment of SRR-TBI.⁵ Activities associated with the SRR-TBI varied by age and gender. There are many factors believed to play a role in how an individual is able to diffuse the impact sustained from a possible concussive force. Musculoskeletal strength as well as the musculoskeletal weakness of the individual—specifically the core and neck musculature, the levels of CSF of the individual, and the self-awareness of either a direct or indirect impact—are all believed to be possible factors in one's ability to defuse a possible concussive hit. For instance, the role of the CSF inside the cranium may play a role in decreasing concussions. If the forces exerted are not great enough to exceed the threshold needed to force

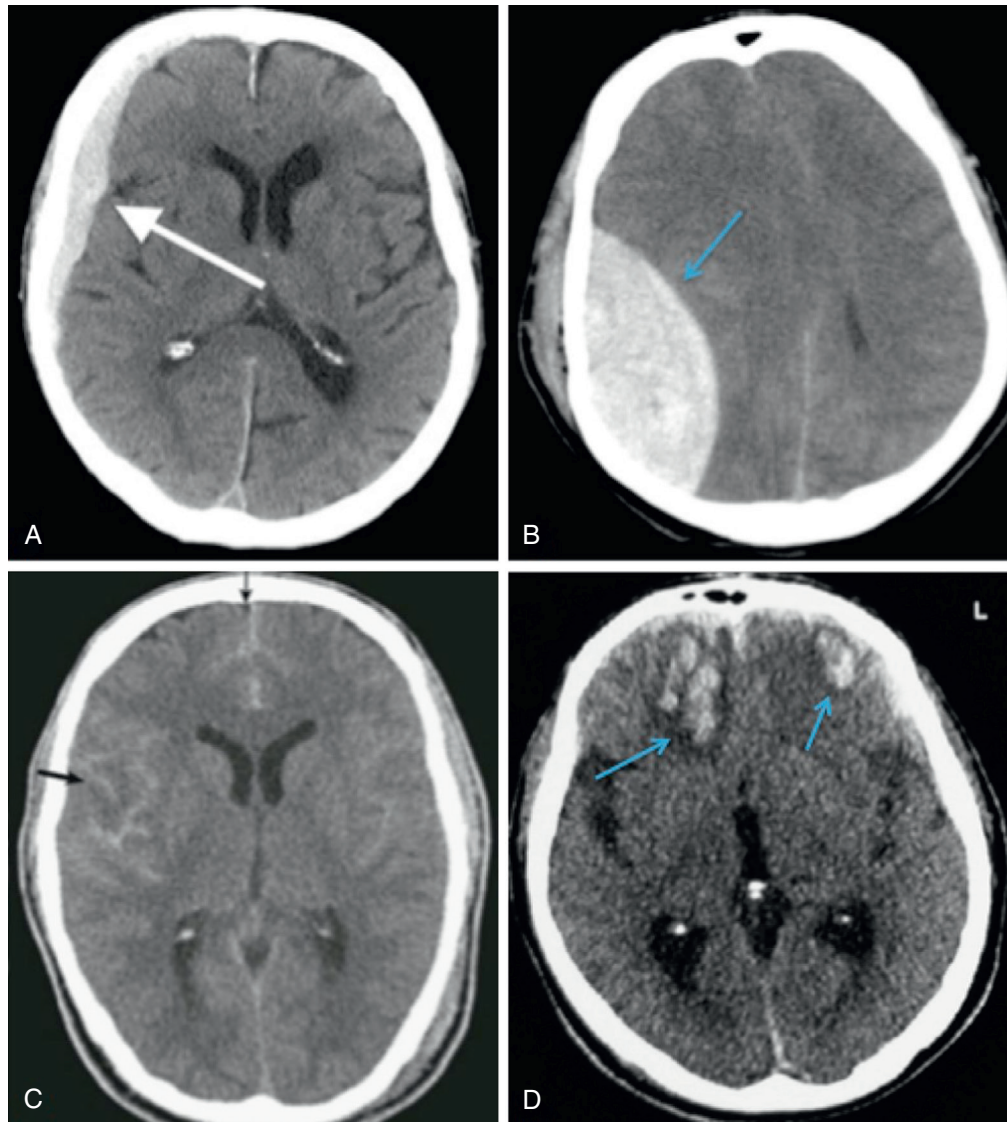


Fig. 122.3 Axial computed tomography scans demonstrating intracranial hemorrhages. (A) Acute right subdural hematoma (*arrow*) following the contour of the brain. Note the concave shape. (B) Acute right epidural hematoma (*arrow*) confined between the coronal and lambdoid sutures. Note the convex shape. (C) Acute traumatic right frontal and temporal subarachnoid hemorrhage (*arrow*). (D) Bilateral frontal basal contusion (*arrow*) where the brain interfaces with the rough base of the anterior cranial fossa.

the brain to impact the interior of the skull, an injury may not occur.⁶ A linear acceleration/deceleration action can occur in multiple different ways. For instance, a top-of-the-head type impact or anterior-to-posterior type injury can cause a coup or a coup-countercoup-type mechanism of injury. A coup-countercoup type of injury occurs on the side of the impact, as well as on the opposite side due to the rebounding of the brain inside the cranium, whereas a coup injury occurs only at the site of impact. Rotational acceleration impacts are thought to be more likely the cause of loss of consciousness in concussions rather than linear/translational acceleration types of impacts.⁶ This is likely due to the location of the injury. It is believed that the rotational acceleration impact causes shearing and tensile strains of the cerebrum about the brainstem. Since one of the functions of the brainstem is consciousness, one could

hypothesize that if the brain stem were affected, consciousness could be as well.

The mechanical etiology of concussions and brain injury, as well as their neurologic consequence, are complex and continue to be researched. Concussions represent a clinical dilemma for sports medicine professionals. Understanding the anatomy, biomechanics, and varied symptoms is essential in furthering the understanding and advancement in the course of their treatment.

SPINE

The adult spine is composed of 33 vertebrae, or bony blocks, which protect the spinal cord and nerve roots, and support the cranium. The typical arrangement is 7 cervical, 12 thoracic,

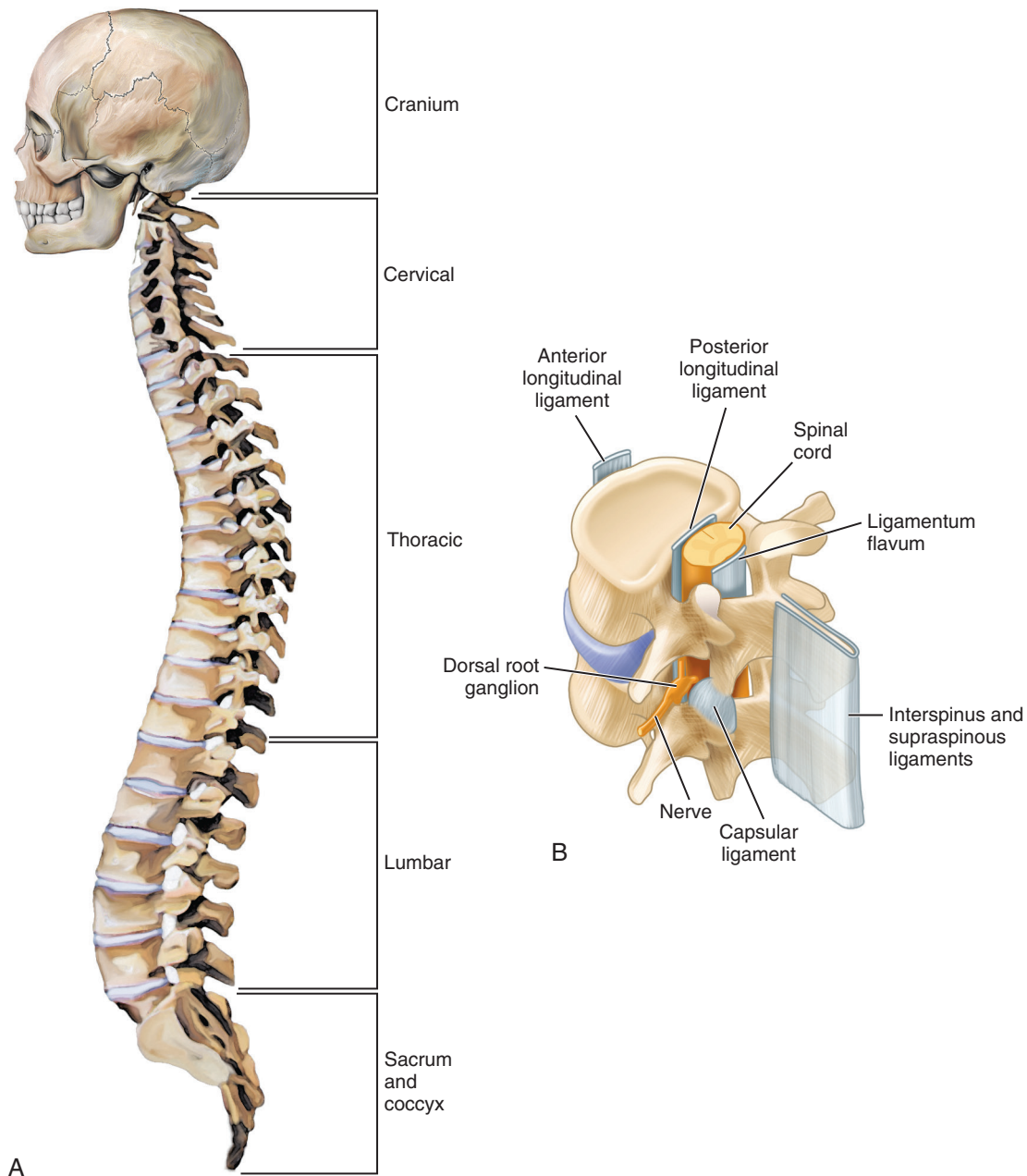


Fig. 122.4 The lateral view of the skull and spine (A). Detailed view of a spinal motion segment consisting of two vertebrae and the intervertebral disk, ligaments, and neural tissues (B).

5 lumbar, and 3 to 5 fused sacral and coccygeal vertebrae (Fig. 122.4A), although numerous variations in normal anatomy exist in up to 10% of the population.⁷ Each vertebra from the second cervical (C2) to the first sacral (S1) is connected anteriorly by an intervertebral disc, and posteriorly by paired diarthrodial (facet) joints in addition to supporting ligamentous structures (see Fig. 122.4B). There is variation in size and alignment of the facet joints when moving from cranial to caudal segments, with size increasing more distally in the thoracolumbar and lumbar segments, and the facet joints being more coronally oriented in the thoracic spine, and more sagittally oriented in the lumbar spine.⁸ The three-column concept of spinal stability as described by Denis remains important

in describing spinal injuries of all mechanisms including those from athletic injuries.⁹ According to Denis, the anterior column is composed of the anterior two-thirds of the vertebral body, disk, and anterior longitudinal ligament; the middle column is composed of the posterior third of the vertebral body, disk, and posterior longitudinal ligament; and the posterior column is composed of the facet joints, facet capsules, ligamentum flavum, and interspinous and supraspinous ligaments. When at least two of the three columns are disrupted, the result is a potentially unstable spine that may require surgical stabilization.

The cervical and lumbar spines are lordotic, while the thoracic spine and sacrum are naturally kyphotic (see Fig. 122.4A). Normal

thoracic kyphosis is 10 to 40 degrees, and normal lumbar lordosis is 40 to 60 degrees, noting that an approximate difference of 30 degrees between thoracic kyphosis and lumbar lordosis will place most patients into good sagittal plane balance.¹⁰

The cervical spine is composed of seven vertebrae, including C1 (the atlas), C2 (the axis), and five subaxial (below C2) vertebrae, which share similar characteristics. The atlas has no vertebral body but consists of a bony ring containing lateral masses that lie between the superior and inferior articular processes, which articulate with the occiput and C2 (axis), respectively. Synovial condylar joints are present at the occiput–C1 articulation, whereas the C2–C3 vertebrae and all subaxial vertebrae are connected by an intervertebral disc anteriorly, and by paired diarthrodial (facet) joints posteriorly, along with supporting ligaments. The intervertebral disc primarily resists axial loads and acts as the “shock absorber” of the spine, while the posterior elements, including the facet joints, provide rotational stability and resist extension forces.¹¹ The second cervical vertebra contains a peg-shaped odontoid process (dens) that articulates with the anterior C1 ring and accounts for approximately 50% of cervical rotation, while the occiput–C1 junction accounts for approximately 50% of flexion and extension.¹² There is a complex interplay of ligaments at the craniocervical and atlantoaxial junctions, which allow for a physiologic range of motion (ROM) of the cervical spine. The anterior longitudinal ligament runs anterior to the vertebral bodies; the posterior longitudinal ligament is contiguous with the tectorial membrane; and the apical and transverse ligaments stabilize the atlantodental and atlantoaxial motion segments, respectively (Fig. 122.5A). One unique feature of the cervical spine is the presence of the uncovertebral joints of Luschka, allowing a bony articulation at the lateral edge of each disc space. The paired facet joints at the posterolateral aspect of each motion segment contain synovial fluid between the articular cartilage segments contained by a ligamentous capsule, and provide stability to torsional and bending loads.¹³ The vertebral artery arises from the subclavian

artery and enters the spine at the transverse foramen at C6 in most cases, providing blood supply to the posterior cerebral circulation via the basilar artery. It can be vulnerable to injury in cervical spine trauma due to its close confines as it passes proximally from the C6 transverse foramen to the C1 foramen (V2 segment) and then passes along the posterior superior C1 ring (V3 segment) until it reaches approximately 1 cm from the midline and then enters the dura (V4 segment).¹⁴ The cervical spine attaches to the thoracic spine and rib cage at the cervicothoracic junction, where the seventh cervical lateral mass articulates with the superior articular facet of the first thoracic vertebra and first rib. The cervical spine is the segment most prone to catastrophic injury in sports as it lacks the inherent stability of the rib cage at the thoracic levels, and contains the cervical spinal cord that is critical for upper and lower extremity motor and sensory function in addition to bowel and bladder control.

The kyphotic thoracic spine contains smaller vertebral bodies than those in the cervical spine, and has facets that are oriented in a more coronal plane. The transverse processes are longer and contain several articulations with the adjacent rib head. Rather than lateral masses containing the superior and inferior articulating facets, as in the cervical spine, the thoracic and lumbar spine contain a pars region, which is a narrow, lateral extension of the lamina that acts as a bridge between the superior articulating and inferior articulating facets. This area is especially prone to injury at the C2 and L5 levels due to regional anatomic variations, (elongation at C2, and thin medial to lateral dimension at L5).¹⁵ In the lumbar spine, the vertebral bodies are larger in order to support the trunk and greater forces in this region, and the facet joints are more sagittally oriented, which allows for more flexion and extension. As noted previously, anatomic variations are present in approximately 10% of spines, commonly consisting of segmentation anomalies (sacralization of L5, rudimentary rib at L1), spina bifida occulta (absence of a portion or all of the lamina) and pars defects.

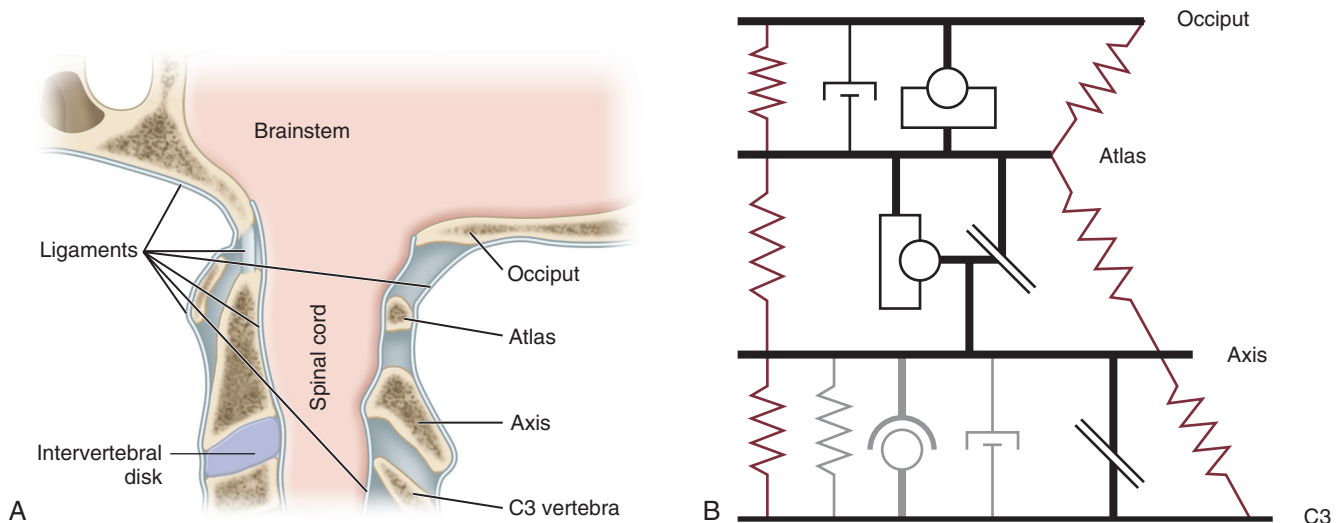


Fig. 122.5 The major anatomic tissues of the upper cervical spine (A) and the matching mechanical analogues for biomechanical modeling using a free-body diagram (B). A ligament can be modeled by a spring; and a disk by a spring and dashpot.

INTERVERTEBRAL DISK

The intervertebral disk is a fibrocartilaginous structure that forms the mobile articulation between adjacent vertebrae in the anterior and middle columns from the C2 to S1 levels. In an adult, the disk is avascular and receives its nutrients through diffusion from the cartilaginous endplates. It displays both viscoelastic and time-dependent properties, and contains a soft, inner core (the nucleus pulposus) and a more rigid, outer ring containing concentric lamellae (annulus fibrosus).^{16,17} The annulus is a ring containing type I collagen arranged in concentric lamellae approximately 65 degrees from the spinal axis, while the nucleus is an amorphous gel of type II collagen and proteoglycans, composed of almost 90% water in a young, healthy person.^{18,19} The outer layers of the annulus fibrosus are innervated by sensory nerve structures, whereas the inner annulus and nucleus lack innervation. This is important in that intervertebral disk herniations can become painful when pathologic or repeated physiologic stresses affect the structural integrity of the annulus.²⁰ The nucleus pulposus is able to distribute and transfer compressive loads through its attachments to the surrounding annulus fibrosus, and loss of this normal physiologic process leads to increased load transfer to the facet joints and posterior structures.²⁰⁻²³ Disc degeneration occurs as water content is lost and annular fissuring occurs; this is most common clinically at the L4–L5 and L5–S1 levels as well as the lower cervical spine.²⁴⁻²⁶

LIGAMENTS

As noted previously, the middle and posterior columns of the spine contain various ligaments that connect each vertebra to the adjacent vertebrae. Although the upper cervical spine has a different and more complex arrangement, the subaxial cervical spine, thoracic and lumbar spines share a common ligamentous structure. This includes the anterior and posterior longitudinal, ligamentum flavum (yellow ligament), interspinous, supraspinous, and bilateral facet capsules with their associated ligaments (see Fig. 122.4B). The upper cervical spine includes ligamentous attachments between the occiput, the atlas (C1) and the axis (C2), including the transverse, apical, alar, and accessory ligaments in addition to the anterior and posterior longitudinal as well as the ligamentum nuchae, a dense fibrous band contiguous with the supraspinous ligament extending from the occipital protuberance to the C7 spinous process. Ligaments throughout the spine are richly innervated and can provide proprioceptive feedback on head and neck position, and assist in stabilizing the cervical spine.^{27,28} Ligaments are frequently involved in sports injuries to the spine given their important role in providing stability, and loss of innervation can result in significant rehabilitation demands on the injured athlete. Generally speaking, spinal ligamentous injuries are slower to heal than bony injuries and can lead to late instability if unrecognized at an early stage. A comprehensive review of ligamentous injury and strain are beyond the scope of this chapter, and more information can be found elsewhere in this edition.

MUSCLES

The majority of body weight is located anterior to the spinal column, so strong muscle groups are necessary to stabilize the spine and allow physiologic movement without compromising structural integrity. There are three basic groups of spinal muscles, the superficial, intermediate (both extrinsic), and deep (intrinsic). The superficial extrinsic muscles include the trapezius, latissimus dorsi, levator scapulae, and rhomboids, and primarily connect the upper limbs to the trunk.¹ The intermediate extrinsic muscles are comprised of the serratus posterior and serve as accessory respiratory muscles. The deep (intrinsic) muscle layer itself includes superficial, intermediate, and deep layers, and plays a role in providing an extensor moment to the spine as well as rotation and side bending, with innervation from the dorsal primary division of the spinal nerves. Lumbar flexion is accomplished primarily by the “core” abdominal muscles consisting of the internal and external obliques in addition to the transversus abdominus. Spinal muscles can be injured creating pain and dysfunction, in addition to loss of important stabilizing forces leading to further injury of vertebrae, ligaments, and discs if not addressed.

NEUROLOGIC TISSUES

The spinal cord and paired nerve roots are the neurologic structures that must be protected by the bones, ligaments, disc, and supporting muscles. The spinal cord originates proximally at its junction with the brainstem at the foramen magnum and ends in most adults at L1 or at the L1–L2 disc space level. The spinal cord ends more distally in children, and conditions such as tethered cord or spine bifida occulta can create aberrant neurologic anatomy. The spinal cord contains a central H-shaped gray matter composed of the cell bodies of interneurons and motor neurons, and a surrounding white matter composed of ascending and descending myelinated tracts (Fig. 122.6). The blood supply of the spinal cord is derived from a median ventral anterior spinal

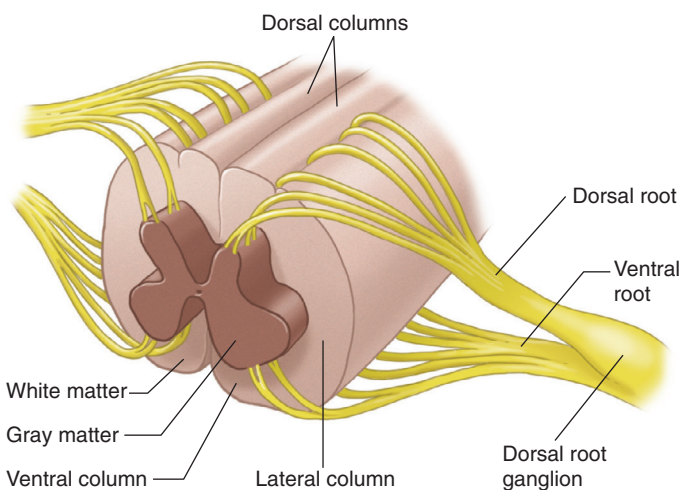


Fig. 122.6 Cross-section of the spinal cord demonstrating the relationship between the gray and white matter, the nerve root, and the dorsal root ganglion.

artery and two smaller dorsolateral spinal arteries.²⁹ Nerve roots exit the spinal canal at each level from the occiput-C1 articulation to the sacrum, and the distal spinal cord (conus medullaris) becomes the cauda equina (“horse’s tail”) distal to L1, which terminates at the S5 level. The spinal cord is surrounded by meninges that include the dura mater, arachnoid mater, and pia mater (moving from superficial to deep, analogous to the meninges surrounding the brain). Both anterior and posterior nerve roots extend from the spinal cord at each level, with the anterior and posterior primary rami forming the spinal nerve as it exits the neural foramen between the pedicles superiorly, the disc anteriorly, and the facet capsules posteriorly (see Fig. 122.4B). The spinal canal is bordered by the posterior vertebral body and disc anteriorly, the pedicles laterally, and the laminae dorsally. The dorsal root ganglion is located within the neural foramen and transmits important somatosensory information. This structure is very sensitive to external compression and is found distal to the dorsal and ventral roots, but proximal to the spinal nerve (see Fig. 122.6).³⁰ Afferent (sensory) fibers generally travel more dorsally within the spinal cord, whereas motor units travel more ventrally, leading to various SCI patterns depending on which anatomic section of the cord is damaged (e.g., anterior cord syndrome leading to motor impairment from ventral pathology including anterior spinal artery compromise). Medial branches of the dorsal primary ramus innervate the facet joints and are sometimes implicated in facet-mediated pain syndromes, while the disc is innervated posteriorly by the sinuvertebral nerve as discussed previously in this chapter.

BIOMECHANICS

As described earlier, the spinal column is a complex mechanical structure with multiple articulations. These articulations permit substantial motion in multiple planes both at the individual functional spinal unit and on a more regional basis. The biomechanical performance of the spine depends on several factors, including the material and constitutive properties of the involved tissues, the magnitude and direction of the applied loads, the rate and duration of applied loading, and the position of the head/spine at the time of loading.³¹ The study of spinal biomechanics has utilized *in vivo* and cadaveric models as well as computational analysis to understand both the normal and the injured conditions.

NORMAL/PHYSIOLOGIC BIOMECHANICS

The unique anatomy of the craniocervical junction (occiput-atlas-axis) results in distinct biomechanics from the remainder of the cervical spine (see Fig. 122.5A). The occipital-axial complex can rotate up to 50 degrees in flexion-extension and 90 degrees in axial torsion, which represents approximately 50% of the motion of the entire cervical spine in these directions. However, the upper cervical anatomy permits only limited lateral bending (± 10 degrees). The subaxial levels are therefore the main contributors to the total cervical lateral bending range (20 to 45 degrees).^{32,33} On a segmental basis, the ROM in the subaxial cervical region is similar at each level ranging from 4 to 23

degrees of sagittal plane motion.^{34,35} Due to the complex anatomic shape of the vertebrae, movement of the functional spinal unit along one (primary) axis is accompanied by movement along a coupled (secondary) axis. This coupled motion is commonly cited in the subaxial cervical spine, where lateral bending occurs concomitantly with axial rotation in the same direction due to the orientation of the facet articulations.³⁶ The stabilizing effect of the ribcage limits the motion of the thoracic spine. The coronally oriented facets further limit flexion-extension. The upper thoracic spine displays coupled motion in lateral bending and axial rotation, whereas coupling in the lower thoracic spine is less pronounced.³⁷ The more sagittally oriented facets of the lumbar spine permit greater flexion-extension than the thoracic region, averaging 12 degrees at L1–L2 up to 17 degrees at L5–S1.

Despite regional differences, the general biomechanical responses of the subaxial (C2–C7) cervical, thoracic, and lumbar regions are similar. The intervertebral disk, ligaments, and bilateral facet joints limit and guide the relative motions of adjacent vertebrae, which leads to highly nonlinear load-displacement relationships (Fig. 122.7).³⁸ These relationships are characterized by curves featuring two main zones: the neutral zone and the elastic zone. These two zones together constitute the ROM of a vertebral segment (see Fig. 122.7).³⁸ In the neutral zone, the vertebral motion segment is highly flexible, permitting a great amount of motion for very small applied loads. In the elastic zone, the spinal tissues begin to individually engage and undergo loading, with increasing stiffness as load increases. Such a stiffening behavior results from the viscoelastic properties of the intervertebral disk and of the ligaments as they deform inhomogeneously and nonlinearly with an applied load. The disk absorbs and distributes a compressive load and is mechanically comparable to a spring in parallel with a viscous damper (see Fig. 122.5B). Ligaments are like cables and can only resist tensile loading; as such, the posterior spinal ligaments tighten during spinal flexion and the anterior ligaments tighten during extension. These tissues are mechanically analogous to springs (see

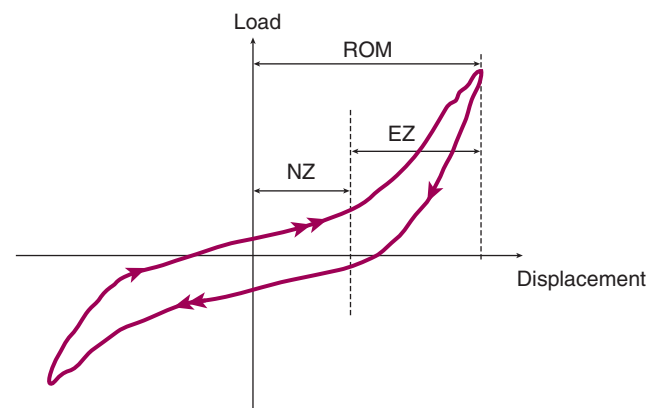


Fig. 122.7 A representative load-displacement curve for a motion segment showing the nonlinear relationship between the segmental displacement (translation or rotation) and load (force or moment) generated. The difference between the loading (*double arrow*) and unloading (*single arrow*) indicates hysteresis. The elastic (EZ) and neutral (NZ) zones are defined, and together comprise the total range of motion (ROM).

Fig. 122.5B). The cervical neural arch and the facet joints also transfer and support 3% to 25% of a compressive load and limit the motion between adjacent vertebrae, especially in the extremes of extension and anterior translation.³⁹⁻⁴¹ Although the orientation of the cervical facet joints, with respect to both the coronal and sagittal planes, increase moving caudally, the ROM in the subaxial cervical spine is similar at each level, ranging from 4 to 23 degrees in human voluntary and cadaveric studies of sagittal bending.^{32,34,35}

As the primary axial stabilizing structure of the head and trunk, the spinal column is subjected to large loads during daily activities. The vertebral bodies can bear compressive forces of up to 1700 N, 4000 N, and 8000 N in the cervical, thoracic, and lumbar regions, respectively.³³ Not surprisingly, the lumbar vertebrae, which are required to sustain more of the body's weight, can support a greater axial load. Because of the different loading patterns and vertebral structures, the spinal ligaments also exhibit a wide range of failure strengths that range from 76 to 436 N in the upper cervical spine, with the cruciate, alar, and anterior longitudinal ligaments being the strongest. In the subaxial cervical spine, the capsular ligaments and ligamentum flavum are the strongest, with failure strengths ranging between 47 and 264 N. The anterior and posterior longitudinal ligaments in the lumbar spine are the strongest, with failure strengths reported from 100 to 510 N. Similarly, the maximum compressive load that the intervertebral disk can withstand also increases from the cervical to lumbar spines: 74, 1800, and 5300 N in the cervical, thoracic, and lumbar regions, respectively. Intervertebral disks are weaker in shear and can only resist shear loads of 20 N in the cervical spine and 150 N in the thoracic region. Without protection from the facet joints, the intervertebral disks can only resist 1.8 and 31 Nm of torsion in the cervical and lumbar spines, respectively.³³ While unaccounted in most cadaveric biomechanical studies, muscle activation not only stabilizes the spine and helps to protect it from excessive motion, but also it can produce loading to individual ligaments beyond their threshold for injury.^{42,43}

The anatomic structures of the spine can fail when mechanical loading is outside a physiologic range. Injurious loads can result from a variety of scenarios: the combination of different loads, the application of physiologic loads in extreme spinal positions, and/or even from repetitive loading (fatigue) within the normal range.⁴⁴⁻⁴⁷ The ensuing structural failures of the vertebrae, disk, and/or ligaments depends on the magnitude of the loads, their direction, and the rate of loading. For example, the neutrally aligned cervical spine is lordotic, and energy inputs can be dissipated to some extent by the intervertebral disks and paraspinal muscles. Numerous biomechanical studies have suggested that with flexion of approximately 30 degrees, the straightened cervical spine assumes the physical characteristics of a segmented column. Axially applied force is directly transmitted to the spinal structures. When the threshold vertical compression is reached, the cervical spine fails and buckles in a flexion mode with anterior column failure and possible facet subluxation or dislocation.⁴⁸⁻⁵⁰

Other examples of this phenomenon are the injuries sustained by quarterbacks or rugby players when they are sacked from

behind. At a high rate of loading, the instantaneous axis of rotation (IAR) of a vertebra relative to its subadjacent vertebra moves upward into the top moving vertebra from its normal location during physiologic motions under the intervertebral disk. As a consequence, the inferior facets of the superior vertebra have an impact on the superior facets of the inferior vertebra instead of gliding posteriorly on them.^{51,52} Such an extreme compression can produce facet joint hemarthrosis, cartilage fissures, and even facet fracture.^{53,54} Given the complementary stabilizing effects of the discoligamentous structures, isolated injury may result in varying degrees of instability. For instance, in a cadaveric study, both the ROM and neutral zone have been shown to significantly increase by 50% to 60% in flexion after transection of the spinal ligaments (except the posterior longitudinal ligament) at the C5–C6 level.⁵⁵ In contrast, an annular cut of the C5–C6 intervertebral disc does not destabilize the segment until after facet capsule transection.⁵⁵ Similarly, a unilateral locked C5–C6 facet joint reduces segmental ROM by up to 3.6 degrees in all directions except ipsilateral torsion.⁵⁶ Major damage and failure of the osseous and ligamentous spinal tissues can result in serious neurologic pathologies.^{57,58} Because the spinal cord and neural elements are no longer protected in such cases, they can be subjected to injurious loading (e.g., compression and shear) by the surrounding tissues. However, subtle and undetected structural failures of tissue can also occur and can lead to acute and chronic neurologic pathologies. Minor tissue damage can render the spine unstable and lead to nonphysiologic loading of the neurologic tissues in particular locations. For instance, subfailure stretching of the cervical capsular ligament in experimental *in vivo* animal studies has been shown to produce microstructural collagen fiber realignment and laxity, which can lead to central sensitization.^{59,60} In addition, neurologic and vascular tissues can be loaded under nonphysiologic conditions even when the soft tissues of the spine only sustain minor damage. For instance, a transient reduction of the spinal canal diameter is induced by altered vertebral kinematics during high loading rates and can lead to spinal cord contusion.⁵¹ While the unique architecture of the intervertebral disk enables it to act as a major stabilizing structure of the spine, degenerative changes may have significant effects on this function. In a cadaveric study, lumbar motion segments with increasing degrees of disc degeneration were found to have erratic excursions of the IAR, vertical translation of the center of rotation, and abnormal distribution across the disc space with lower pressures in the region of the nucleus and pressure spikes in the annular region.⁶¹ The result is an unstable functional spinal unit and potentially increased risk of flare ups of back pain with routine daily or sporting activities.

Integration of Biomechanical Considerations for Treatment

The application of spinal biomechanics to the context of sports injury remains challenging, despite an abundance of research in this field. While some broad injury mechanisms, such as axial load, torsion, flexion, and extension, can be applied across all sports injuries to the spinal column, there is a great variability between how these injuries occur among different sporting activities. This is further complicated by the age of the athlete and

any pre-existing degenerative changes that are present; the ability of the surrounding musculature to support or withstand non-physiologic forces; and the individual athlete's prior exposure or injuries.

As discussed previously, the function of the human spine is to support body weight, protect the neural elements, and to maintain posture as well as resist external forces. Despite the ability of the spine to function well under a variety of loading scenarios, athletic activities can cause damage through either repetitive physiologic loading mechanisms or through load to failure if enough force is applied in a specific manner, such as catastrophic cervical SCI through tackling in football.

The regions of the spine most vulnerable to injury generally include the cervicothoracic and thoracolumbar junctions (transitional zones between the mobile cervical and lumbar spines, and the rigid thoracic spine and accompanying rib cage), although any spinal level can sustain injury if the ability of the supporting structures in that area are compromised. Spine injury patterns in sports remain sport-specific to a large degree, and this information can be found elsewhere in this text in further detail.

SUMMARY

The spinal column is composed of bone, disc, ligaments, and the surrounding muscles that serve to protect the neural elements and maintain posture as well as to resist and respond to various external loads. TBI and spinal cord injuries are among the most feared of all sports injuries due to their potential catastrophic and permanent nature. A greater understanding of spinal biomechanics has been helpful in revising the rules of play in various sports (e.g., the banning of spear tackling in American football), and in improving the design of protective equipment. Despite the use of the protective equipment, the cervical spine remains the most vulnerable to injury, and these injuries often have the most devastating consequences.

The CNS has limited healing potential once an injury occurs; therefore prevention is the best method of minimizing brain and spine injuries in sports. As the pathologic mechanisms associated with spine and brain injuries in sports remain incompletely understood, further research is needed to minimize their effects, especially in the realm of SCI. In rare cases, neurologic injury can occur despite no obvious bony or ligamentous injury, and observation of the head and trunk position to determine the mechanism of injury does not always offer useful information. All on-field personnel should be well trained to handle brain

and spine injuries and to assume the worst-case scenario (an unstable spine) when caring for or transporting the athlete off the field.

For a complete list of references, go to [ExpertConsult.com](https://www.expertconsult.com).

SELECTED READING

Citation:

Denis F. The three-column spine and its significance in the classification of acute thoracolumbar spinal injuries. *Spine*. 1983;8(8):817–831.

Level of Evidence:

IV

Summary:

Retrospective study of 412 thoracolumbar injuries, defining the concept of an anterior, middle, and posterior column osteoligamentous complex injuries. Mechanistic descriptions are applied to four basic injury patterns forming the basis for modern spinal injury classification systems.

Citation:

Panjabi MM. The stabilizing system of the spine. Part II. Neutral zone and instability hypothesis. *J Spinal Disord*. 1992;5(4):390–397.

Level of Evidence:

V, review

Summary:

Elegant hypothesis of the concept of spinal stability as it relates to passive and active stabilizers, normal (physiologic) function of the spinal motion segment, and pathologic spinal motion leading to dysfunction or pain. Neural and muscular adaptive mechanisms are discussed.

Citation:

Jaumard NV, Welch WC, Winkelstein BA. Spinal facet joint biomechanics and mechanotransduction in normal, injury and degenerative conditions. *J Biomech Eng*. 2011;133(7):071010.

Level of Evidence:

V, review

Summary:

Thorough review of the zygapophyseal (facet) joint anatomy and biomechanics with a focus on differences between normal, injury, and degenerative states. The authors provide a review of the biomechanical data on capsular strain, the response to load and mechanotransduction, and the use of finite element models.

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